

Simultaneous equations and inequalities - Set 1 - Solutions

Specification MM1.4, MM1.5, M4.16, M4.17.

Answer key

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Worked solutions

Question 1 SIM-02 · Simultaneous equations and inequalities · spec MM1.5

Find the complete set of values of x for which $2x^2 - 5x - 3 \geq 0$.

Answer: A

$(2x + 1)(x - 3) \geq 0$: the roots are $-\frac{1}{2}$ and 3 and the parabola opens upwards, so $x \leq -\frac{1}{2}$ or $x \geq 3$.

Full method - how and why

A quadratic inequality with a leading coefficient other than 1. Factorise, find the roots, then use the shape of the parabola.

1 Factorise: $2x^2 - 5x - 3 = (2x + 1)(x - 3)$.

why: Two numbers multiplying to $2 \times (-3) = -6$ and adding to -5 : -6 and 1 , so split $-5x$ as $-6x + x$.

2 Roots: $x = -\frac{1}{2}$ and $x = 3$.

why: Set each bracket to zero.

3 The coefficient of x^2 is positive, so the parabola opens upwards and is ≥ 0 **outside** the roots.

why: A quick sketch is the safest way to convert a factorisation into an inequality answer.

4 Answer: $x \leq -\frac{1}{2}$ or $x \geq 3$, with $\leq$ because equality is allowed.

why: The question has ≥ 0 , so the roots themselves are included.

Where it goes wrong

- Giving $-\frac{1}{2} \leq x \leq 3$, which is where the expression is negative.
- Mixing up the roots as -3 and $\frac{1}{2}$ - the root of $(2x + 1)$ is $-\frac{1}{2}$, not -2 .

Takeaway Factorise, mark the roots, sketch the parabola: 'outside' for > 0 , 'between' for < 0 when $a > 0$.

Question 2 SIM-03 · Simultaneous equations and inequalities · spec MM1.5

Find the set of values of x satisfying both $3x - 7 < 2$ and $x^2 > 4$.

Answer: A

$3x < 9 \Rightarrow x < 3$. $x^2 > 4 \Rightarrow x < -2$ or $x > 2$. Combining: $x < -2$ or $2 < x < 3$.

Full method - how and why

Two inequalities joined by 'and'. Solve separately, then take the overlap - the second one produces two intervals, which is where the care is needed.

1 Solve the linear one: $3x - 7 < 2 \Rightarrow 3x < 9 \Rightarrow x < 3$.

why: Dividing by a positive number leaves the inequality direction unchanged.

2 Solve the quadratic one: $x^2 > 4 \Rightarrow |x| > 2 \Rightarrow x < -2$ or $x > 2$.

why: $x^2 > 4$ is satisfied by large positive **and** large negative x - taking only the positive branch is the classic error.

3 Intersect the sets on a number line: from $x < 3$, keep the parts of $x < -2$ (all of it) and of $x > 2$ (only up to 3).

why: 'Both' means intersection; drawing both solution sets on a single line prevents mistakes.

4 Answer: $x < -2$ or $2 < x < 3$.

why: The answer is itself two intervals - a perfectly legitimate solution set.

Where it goes wrong

- Dropping the branch $x < -2$ and answering $2 < x < 3$.
- Solving $x^2 > 4$ as $x > \pm 2$, which is not a meaningful statement.

Takeaway $x^2 > k$ gives two intervals; intersect solution sets on a number line rather than in your head.

Question 3 SIM-04 · Simultaneous equations and inequalities · spec MM1.4

The line $y = x + k$ meets the circle $x^2 + y^2 = 8$ at exactly one point. Find the possible values of k .

Answer: C

$$x^2 + (x + k)^2 = 8 \Rightarrow 2x^2 + 2kx + k^2 - 8 = 0. \text{ Tangent when } 4k^2 - 8(k^2 - 8) = 0 \Rightarrow -4k^2 + 64 = 0 \Rightarrow k = \pm 4.$$

Full method - how and why

A line meeting a circle exactly once is a tangency condition: substitute and set the discriminant to zero.

1 Substitute $y = x + k$ into $x^2 + y^2 = 8$: $x^2 + (x + k)^2 = 8$.

why: Eliminate y so the intersections are the roots of a quadratic in x .

2 Expand and collect: $2x^2 + 2kx + (k^2 - 8) = 0$.

why: $(x + k)^2 = x^2 + 2kx + k^2$, so the x^2 terms combine to $2x^2$.

3 Exactly one point means one repeated root: discriminant $= 0$, so $(2k)^2 - 4(2)(k^2 - 8) = 0$.

why: Tangency is precisely the repeated-root case.

4 Simplify: $4k^2 - 8k^2 + 64 = 0 \Rightarrow -4k^2 = -64 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$.

why: Two tangent lines exist - one on each side of the circle - so both signs are genuine.

Where it goes wrong

- Answering $k = \pm 2\sqrt{2}$ by confusing the radius $\sqrt{8}$ with the intercept.
- Giving only $k = 4$; the geometry guarantees a matching tangent with $k = -4$.

Takeaway Tangent to a circle: substitute and set the discriminant to zero - or use 'distance from centre = radius', which gives $\frac{|k|}{\sqrt{2}} = 2\sqrt{2}$ just as fast.

Question 4 SIM-05 · Simultaneous equations and inequalities · spec MM1.5

For how many integers n is $n^2 - 10n + 16 < 0$?

Answer: A

$$(n - 2)(n - 8) < 0 \Rightarrow 2 < n < 8, \text{ so } n = 3, 4, 5, 6, 7: \text{ five integers.}$$

Full method - how and why

Counting integers in the solution set of a quadratic inequality. The counting at the end is where the marks are won or lost.

1 Factorise: $n^2 - 10n + 16 = (n - 2)(n - 8)$.

why: Two numbers multiplying to 16, adding to -10 : -2 and -8 .

2 Negative between the roots: $2 < n < 8$.

why: Upward parabola, so it dips below the axis strictly between its roots.

3 List the integers strictly inside: 3, 4, 5, 6, 7 - five values.

why: Strict inequality excludes 2 and 8, where the expression equals zero.

Where it goes wrong

- Including the endpoints and answering 7.
- Counting $8 - 2 = 6$ as the number of integers - the count between a and b exclusive is $b - a - 1$.

Takeaway Write the integers out; the number strictly between a and b is $b - a - 1$, not $b - a$.

Question 5 SIM-06 · Simultaneous equations and inequalities · spec MM1.4

Solve $xy = 6$ and $x - y = 1$. What is the sum of all possible values of $x + y$?

Answer: F

$x = y + 1$: $y^2 + y - 6 = 0 \Rightarrow (y + 3)(y - 2) = 0$. Solutions $(3, 2)$ with $x + y = 5$ and $(-2, -3)$ with $x + y = -5$. The sum is 0.

Full method - how and why

A product and a difference. Substituting the linear relation gives a quadratic with two solution pairs, and the question wants a total over both.

- From $x - y = 1$, write $x = y + 1$.
why: Substituting the linear equation keeps things to one variable.
- Substitute into $xy = 6$: $(y + 1)y = 6 \Rightarrow y^2 + y - 6 = 0$.
why: Expand before rearranging into standard form.
- Factorise: $(y + 3)(y - 2) = 0$, so $y = -3$ or $y = 2$.
why: Both are valid - nothing in the question restricts the sign.
- Find the pairs: $y = 2 \Rightarrow x = 3$, giving $x + y = 5$; $y = -3 \Rightarrow x = -2$, giving $x + y = -5$. Sum of these: 0.
why: The question asks for the sum of **all** possible values of $x + y$, so compute $x + y$ for each pair and add.

Where it goes wrong

- Finding only the positive pair and answering 5.
- Adding all four numbers $3 + 2 + (-2) + (-3)$ - which happens to be 0 too, but for the wrong reason; on another question that luck runs out.

Takeaway When a system has several solution pairs, compute the requested quantity for each pair and then combine.

Question 6 SIM-07 · Simultaneous equations and inequalities · spec MM1.5

The inequality $x^2 + bx + 9 > 0$ holds for all real x . Find the complete set of values of b .

Answer: D

Need discriminant negative: $b^2 - 36 < 0$, so $-6 < b < 6$.

Full method - how and why

'True for all x ' for an upward parabola means it never touches the axis: discriminant strictly negative.

- Note the leading coefficient is $1 > 0$, so the parabola opens upwards.
why: For > 0 everywhere you need both the right shape and no real roots; here the shape is automatic.
- Require no real roots: $b^2 - 4(1)(9) < 0$.
why: If there were real roots the curve would touch or cross the axis and the expression would fail to be positive there.
- Solve: $b^2 < 36 \Rightarrow -6 < b < 6$.
why: $b^2 < 36$ means $|b| < 6$, a single interval.

Where it goes wrong

- Solving $b^2 < 36$ as two outer intervals - that is the solution of $b^2 > 36$.
- Using $\leq$: at $b = \pm 6$ the curve touches the axis, where the expression equals 0 and is not > 0 .

Takeaway 'Positive for all x ' $\Leftrightarrow a > 0$ **and** $b^2 - 4ac < 0$; $x^2 < k$ gives one interval, $x^2 > k$ gives two.

Question 7 SIM-09 · Simultaneous equations and inequalities · spec MM1.4

The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .

Answer: C

$$x^2 - 2x + (c - 1) = 0 \text{ has one solution when } 4 - 4(c - 1) = 0, \text{ so } c = 2.$$

Full method - how and why

'Exactly one solution' for a line and a parabola is again the discriminant-zero (tangency) condition.

- 1 Set the expressions equal: $x^2 + 2x + c = 4x + 1$.

why: A simultaneous solution is a point where both y-values agree.

- 2 Collect: $x^2 - 2x + (c - 1) = 0$.

why: Move everything to one side; the $2x - 4x$ gives $-2x$.

- 3 Discriminant zero: $(-2)^2 - 4(1)(c - 1) = 0$.

why: One solution means a repeated root.

- 4 Solve: $4 - 4c + 4 = 0 \Rightarrow 4c = 8 \Rightarrow c = 2$.

why: Careful with $-4(c - 1) = -4c + 4$.

Where it goes wrong

- Mishandling $-4(c - 1)$ and getting $c = 0$.
- Setting the discriminant > 0 ('has a solution') rather than $= 0$ ('exactly one').

Takeaway Exactly one intersection between a line and a parabola always means discriminant $= 0$.

Question 8 SIM-12 · Simultaneous equations and inequalities · spec MM1.4

The curves $y = x^2 - 4$ and $y = -x^2 + 2x + k$ intersect at exactly two points. Find the complete set of values of k .

Answer: C

$$2x^2 - 2x - 4 - k = 0; \text{ discriminant } 4 + 8(4 + k) > 0 \Rightarrow 36 + 8k > 0 \Rightarrow k > -\frac{9}{2}.$$

Full method - how and why

Two parabolas meeting twice. Subtracting gives one quadratic, and 'two distinct points' is a positive discriminant.

- 1 Set them equal: $x^2 - 4 = -x^2 + 2x + k$.

why: Intersections satisfy both equations, so their right-hand sides agree.

- 2 Collect everything on one side: $2x^2 - 2x - 4 - k = 0$.

why: Note the $-x^2$ becomes $+x^2$ when moved across, giving $2x^2$.

- 3 Discriminant > 0 : $(-2)^2 - 4(2)(-4 - k) > 0$.

why: Here $c = -4 - k$; keeping the whole thing in brackets avoids sign errors.

- 4 Simplify: $4 + 32 + 8k > 0 \Rightarrow 8k > -36 \Rightarrow k > -\frac{9}{2}$.

why: $-4 \times 2 \times (-4 - k) = 32 + 8k$ - two negatives make a positive.

Where it goes wrong

- Losing a sign in $c = -4 - k$ and getting $k > \frac{9}{2}$ or $k > -4$, both of which are offered.
- Forgetting to combine x^2 with $-(-x^2)$ and working with x^2 instead of $2x^2$.

Takeaway Bring both curves to one side before using the discriminant, and bracket the constant term to keep signs straight.

Question 9 SIM-14 · Simultaneous equations and inequalities · spec MM1.4

Solve $2x + 3y = 12$ and $x^2 - y^2 = 0$. How many solutions (x, y) are there?

Answer: B

$x^2 = y^2$ means $y = x$ or $y = -x$. $y = x$: $5x = 12$, one solution. $y = -x$: $-x = 12$, one solution. Two in total.

Full method - how and why

$x^2 - y^2 = 0$ factorises into two lines. The system is a line meeting a pair of lines, so count the intersections of each case.

- Factorise: $x^2 - y^2 = (x - y)(x + y) = 0$, so $y = x$ or $y = -x$.
why: A product is zero when either factor is zero - this splits the problem into two straight-line cases.
- Case $y = x$: $2x + 3x = 12 \Rightarrow 5x = 12$, one solution $(\frac{12}{5}, \frac{12}{5})$.
why: Substitute into the linear equation; a linear equation in one unknown has exactly one root.
- Case $y = -x$: $2x - 3x = 12 \Rightarrow -x = 12$, one solution $(-12, 12)$.
why: Same method; the coefficient of x is non-zero so again exactly one solution.
- Total: two solutions.
why: The two cases give different points, so no double counting.

Where it goes wrong

- Taking the square root as $y = x$ only, halving the answer.
- Concluding 'a line meets a quadratic in at most 2 points, so 2' without checking that neither case is parallel or degenerate.

Takeaway $A^2 = B^2$ means $A = B$ or $A = -B$; both branches must be followed.

Question 10 SIM-17 · Simultaneous equations and inequalities · spec MM1.4

Two numbers have sum 10 and the sum of their squares is 58. What is the positive difference between them?

Answer: B

$(x - y)^2 = 2(x^2 + y^2) - (x + y)^2 = 116 - 100 = 16$, so the difference is 4 (the numbers are 3 and 7).

Full method - how and why

Symmetric information again: a sum and a sum of squares. The identity for $(x - y)^2$ gets the difference without finding the numbers.

- Use $(x - y)^2 = x^2 + y^2 - 2xy$ and $(x + y)^2 = x^2 + y^2 + 2xy$.
why: Adding these gives $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$ - the relation that links all three given quantities.
- Rearrange: $(x - y)^2 = 2(x^2 + y^2) - (x + y)^2$.
why: This avoids having to find xy as an intermediate step.
- Substitute: $2(58) - 10^2 = 116 - 100 = 16$.
why: Careful: it is $2(x^2 + y^2)$, not $x^2 + y^2$.
- Take the positive square root: the difference is 4 (the numbers are 3 and 7).
why: The question asks for the **positive** difference, so take +4.

Where it goes wrong

- Forgetting the factor 2 and getting $(x - y)^2 = -42$, which is impossible.
- Giving 16 (the square of the difference) as the answer.

Takeaway $(x + y)^2 + (x - y)^2 = 2(x^2 + y^2)$: any two of sum, difference and sum-of-squares give the third.

Question 11 X2-14 · Simultaneous equations and inequalities · spec MM1.5

x is a real number. P : ' $x^2 < 4$ '. Q : ' $x < 2$ '. Which one of the following is true?

Answer: C

$x^2 < 4$ means $-2 < x < 2$, which implies $x < 2$ (sufficient); but $x = -5$ satisfies Q and not P (not necessary).

Full method - how and why

Solve the quadratic inequality properly, then compare the two solution sets as intervals.

- 1 $x^2 < 4$ means $|x| < 2$, i.e. $-2 < x < 2$.

why: A quadratic inequality of this shape gives an interval **between** the roots.

- 2 Every x in $(-2, 2)$ satisfies $x < 2$, so $P \Rightarrow Q$: **sufficient**.

why: $(-2, 2)$ is contained in $(-\infty, 2)$.

- 3 Is it necessary? Take $x = -5$: $x < 2$ holds but $x^2 = 25 < 4$. So $Q \Rightarrow P$: **not necessary**.

why: The larger set contains points outside the smaller one.

- 4 So P is sufficient but not necessary.

why: Whenever one solution set sits strictly inside another, the inner condition is sufficient but not necessary.

Where it goes wrong

- Solving $x^2 < 4$ as $x < 2$ and concluding the conditions are equivalent - that drops the lower bound.
- Solving it as $x < -2$ or $x > 2$, which is the solution of $x^2 > 4$.

Takeaway Compare conditions as **sets**: a subset is sufficient for the superset, the superset necessary for the subset.

Question 12 SIM-01 · Simultaneous equations and inequalities · spec MM1.4

Solve the simultaneous equations $x + 2y = 7$ and $x^2 + y^2 = 10$. Find the sum of the possible values of x .

Answer: D

$x = 7 - 2y$: $(7 - 2y)^2 + y^2 = 10 \Rightarrow 5y^2 - 28y + 39 = 0 \Rightarrow (5y - 13)(y - 3) = 0$. $y = 3 \Rightarrow x = 1$; $y = \frac{13}{5} \Rightarrow x = \frac{9}{5}$. Sum of x values $= 1 + \frac{9}{5} = \frac{14}{5}$.

Full method - how and why

One linear and one quadratic equation: substitution is the only method on the specification. The question then asks for the sum of the x -values, so both solutions matter.

- 1 Make x the subject of the linear equation: $x = 7 - 2y$.

why: Substitute the **linear** one into the quadratic, never the other way round - it keeps the algebra to one variable and one squaring.

- 2 Substitute: $(7 - 2y)^2 + y^2 = 10$.

why: Now a single quadratic in y .

- 3 Expand and tidy: $49 - 28y + 4y^2 + y^2 = 10 \Rightarrow 5y^2 - 28y + 39 = 0$.

why: $(7 - 2y)^2 = 49 - 28y + 4y^2$ - the cross term is $2 \times 7 \times (-2y)$.

- 4 Factorise: $(5y - 13)(y - 3) = 0$, so $y = 3$ or $y = \frac{13}{5}$.

why: Check by expanding mentally: $5y^2 - 15y - 13y + 39$. If factorising is awkward, use the formula.

- 5 Back-substitute for x : $y = 3 \Rightarrow x = 1$; $y = \frac{13}{5} \Rightarrow x = 7 - \frac{26}{5} = \frac{9}{5}$. Sum $= 1 + \frac{9}{5} = \frac{14}{5}$.

why: Always return to the **linear** equation for the second variable - it cannot introduce extra solutions.

Where it goes wrong

- Finding only one solution and answering 1 or $\frac{9}{5}$.
- Summing the y -values ($3 + \frac{13}{5} = \frac{28}{5}$) instead of the x -values - that wrong answer is offered.

Takeaway Substitute the linear equation into the quadratic, solve, then back-substitute into the linear equation; read carefully which variable's values are wanted.